

## CHBE – Exam Template 2026

The final exam will be a 90 minutes long written test made of **70 questions**, on all the three modules, divided as follows:

- Cell Biology 30 questions,
- Histology 24,
- Embryology 12 and
- 4 interdisciplinary questions.

Questions' format:

- multiple choice
- open questions
- Fill in the gap, True or False or Matching,
- Blank histology figures
- interdisciplinary questions of mixed form

### **Interdisciplinary:**

1. Which of the following best explains why epithelial tissues can maintain rapid turnover and effective barrier function?

- A) Epithelial cells exhibit high proliferative capacity due to the presence of adult stem cells located in specific niches, and they are tightly joined by junctional complexes that maintain polarity and barrier integrity.
- B) Epithelial tissues are composed of multinucleated cells that divide asynchronously to maintain homeostasis.
- C) Rapid turnover is sustained by fibroblasts that stimulate epithelial mitosis via collagen production.
- D) Epithelial regeneration relies mainly on apoptosis of surface cells, with minimal need for proliferation.
- E) Epithelial cells function independently of the extracellular matrix, allowing for higher renewal rates.

2. Which of the following statements are True or False?

- A) Cardiac muscle cells have high mitochondrial density to sustain continuous ATP production.
- B) Transmission electron microscopy can be used to visualize mitochondrial cristae and matrix density in situ.
- C) Mitochondria in early embryos are maternally inherited and essential for energy supply during cleavage.
- D) Mitochondria cannot be detected by any histological staining techniques.

3. CRISPR-Cas9 genome editing has recently been applied to treat certain hematological diseases. Choose one such condition and briefly explain how CRISPR-Cas9 is used to address it. In your short answer (max 200 words) refer to:

- Cell biology concepts such as gene expression, stem cell manipulation, or DNA repair mechanisms;
- Histological aspects including the structure and function of the affected blood cells or tissues

### **Cell bio module**

4. Briefly describe the structure of ribosomes, their function and localization in cells
5. Which protein gives shape and stability to the nuclear envelope?
- A. Lamin
  - B. Desmin
  - C. Tubulin
  - D. Actin
  - E. Importin

### **Histology module**

6. Identify the INCORRECT statement:
- A. During development, cartilage grows by interstitial and appositional mechanisms
  - B. Volkman's canals connect osteocyte lacunae with haversian canals
  - C. the osteoid is the unmineralized, organic portion of the bone matrix that forms prior the maturation of bone tissue
  - D. The periosteum and endosteum contain osteoprogenitor cells
  - E. The degradation activity of osteoclasts is stimulated by parathyroid hormone and inhibited by calcitonin
7. Which cell type is responsible for the production and secretion of insulin in the pancreas?
- A. Beta cell
  - B. Acinar cell
  - C. Islet cell
  - D. Alpha cell
  - E. Delta cell

### **Embryology module**

8. The part of the sperm containing proteolytic enzymes to digest the zona pellucida is called:
- A) Acrosome
  - B) Capacitor
  - C) Head
  - D) Cumulus
  - E) Corona
9. A 55-year-old man has noted sharp pain in his right lower extremity for the past 2 months. MR imaging of his spine shows impingement on a spinal nerve root by a

herniated structure located between L5 and S1. From which of the following embryonic derivatives does this herniated structure most likely arise?

- A) Notochord
- B) Amnion
- C) Cloaca
- D) Lateral Plate Mesoderm
- E) Neural crest